Begonnen am	Thursday, 15. December 2022, 10:22
Status	Beendet
Beendet am	Thursday, 15. December 2022, 10:48
Verbrauchte Zeit	25 Minuten 45 Sekunden

Frage 1

Vollständig

Erreichbare Punkte: 5,00

What are the differences between “Counterfactual Explanations” and “Adversarial Examples”?

- [ ] a. “Counterfactual Examples” are intuitive explanations for humans
- [ ] b. “Adversarial Examples” are used to interpret a model
- [x] c. “Adversarial Examples” aim to deceive a model
- [ ] d. “Counterfactual Examples” optimize a loss function to generate examples, but “Adversarial Examples” do not optimize a loss function

Frage 2

Vollständig

Erreichbare Punkte: 5,00

The following are valid statements for patterns that may appear in the Time Curves visualization

- [ ] a. patterns can be misleading because of the projection method applied
- [x] b. very similar versions of the same Wikipedia article appear as a cluster pattern
- [x] c. back and forth editing between two versions of the same Wikipedia article appears as an alternation pattern
- [ ] d. big edits/changes of the same Wikipedia article appear as a cluster pattern

Frage 3

Vollständig

Erreichbare Punkte: 5,00

Who are potential users of deep learning visualizations according to the classification by Hohman et al.?

- [x] a. model users
- [ ] b. agents
- [ ] c. data engineers
- [x] d. model developers & builders

Frage 4

Vollständig

Erreichbare Punkte: 5,00

In Permutation Feature Importance

- [ ] a. we compute how much of a model's output variance is explained by the features.
- [x] b. random feature permutation leads to the destruction of information.
- [ ] c. performance on altered datasets is measured.
- [ ] d. features are permuted to lie in a normalized range for comparison.

Frage 5

Vollständig

Erreichbare Punkte: 5,00

Which XAI Methods are usually considered to give **local** explanations?

- [x] a. Saliency Maps
- [ ] b. Concept Bottleneck Models (CBM)
- [ ] c. Testing with Concept Activation Vectors (TCAV)
- [x] d. LIME

Frage 6

Vollständig

Erreichbare Punkte: 5,00

A storytelling article that has multiple predefined substories and where the reader can freely decide which one to follow is known as a

- [x] a. Drill-down story
- [ ] b. Whisky glass structure
- [ ] c. Scrollytelling interface
- [ ] d. Martini glass structure

Frage 7

Vollständig

Erreichbare Punkte: 5,00

The following statements are valid for the Matchmaker visualization technique

- [ ] a. it draws connection bands between heatmaps
- [ ] b. it relies on the continuation grouping principles
- [ ] c. it represents tabular data using a scatterplot matrix
- [x] d. it represents tabular data using a radar chart

Frage 8

Vollständig

Erreichbare Punkte: 5,00

Which of the following statements about LIME are true?

- [x] a. LIME is equivalent to a global surrogate model for small values of the locality parameter.
- [ ] b. The surrogate models used for LIME have to be one-dimensional.
- [x] c. LIME uses local, weighted models fit on randomly sampled data.
- [ ] d. LIME is robust against adversarial attacks.

Frage 9

Vollständig

Erreichbare Punkte: 5,00

The following visualization techniques can typically be used for representing the functioning of sorting algorithms

- [ ] a. attribution maps
- [ ] b. the ConfusionFlow technique
- [ ] c. heatmaps
- [x] d. bump charts

Frage 10

Vollständig

Erreichbare Punkte: 5,00

The following are valid statements for the ConfusionFlow and InstanceFlow techniques

- [ ] a. ConfusionFlow operations on the instance level
- [ ] b. ConfusionFlow shows the per-class confusion as heatmaps
- [x] c. InstanceFlow shows the evolution of items using a Sankey diagram
- [x] d. ConfusionFlow operates on the class level

Frage 11

Vollständig

Erreichbare Punkte: 5,00

The following statements are valid for the data-ink ratio by Edward Tufte

- [ ] a. should be minimized for producing effective visualizations
- [ ] b. should be 0 for producing effective visualizations
- [x] c. should be understood as a general guideline rather than a strict optimization criterion
- [x] d. chart junk decreases the ratio

Frage 12

Vollständig

Erreichbare Punkte: 5,00

The following are examples where machine learning is used for visualization (ML4Vis)

- [x] a. a visualization showing potentially interesting patterns identified by a pre-trained model
- [ ] b. a machine learning model used for detecting usage patterns of a visualization tool
- [ ] c. a machine learning model used for recommending next analysis steps in a visualization tool
- [ ] d. a machine learning model used for suggesting suitable plot types for a given dataset

Frage 13

Vollständig

Erreichbare Punkte: 5,00

Which of the following statements about Network Dissection are true?

- [x] a. Network Dissection is based on comparing regions of strong activation to concept-labeled image segments.
- [ ] b. Network Dissection is a model-specific explanation technique.
- [ ] c. Network Dissection assigns concepts to units.
- [x] d. Network Dissection yields global insights about the network based on activation statistics.

Frage 14

Vollständig

Erreichbare Punkte: 5,00

The following are valid statements for the Time Curves technique

- [x] a. spatial proximity in the visualization indicates similarity between time points
- [x] b. the temporal ordering of the data is preserved
- [ ] c. the layout is independent of the projection method used
- [x] d. can be applied to visualize textual changes in documents

Frage 15

Vollständig

Erreichbare Punkte: 5,00

The following are disadvantages of dimensionality reduction techniques

- [x] a. it can be difficult to preserve the semantics of single attributes/dimensions
- [x] b. the results can be hard to interpret
- [ ] c. outliers in the data cannot be taken into account
- [x] d. the calculation can be time-consuming

Frage 16

Vollständig

Erreichbare Punkte: 5,00

3D is known to be problematic for visualizing multi-dimensional numerical data because of

- [ ] a. animated transitions
- [ ] b. potentially hidden information
- [x] c. perspective distortion effects
- [ ] d. the necessity to rotate the visualization for inspecting the data

Frage 17

Vollständig

Erreichbare Punkte: 5,00

Consider an image for which your neural network predicts a class C. You use GradCAM on the image to get an explanation for class C. You notice that only the background region of the image is highlighted in the GradCAM explanation. Which of the following conclusions can you safely draw from the explanation?

- [ ] a. The background region in the image is similar to that of other images of class C.
- [x] b. The background region is generally important to classify an image as class C.
- [x] c. Changes in the background region are most likely to affect the prediction of class C for most images.
- [ ] d. None of the three conclusions can be safely drawn.

Frage 18

Vollständig

Erreichbare Punkte: 5,00

The following statements are valid in the context of dimensionality reduction techniques

- [ ] a. t-SNE and PCA are linear approaches
- [x] b. t-SNE requires setting hyperparameters
- [x] c. UMAP is computationally faster than t-SNE
- [ ] d. PCA is computationally slower than t-SNE

Frage 19

Vollständig

Erreichbare Punkte: 5,00

The following statements are valid for particle-based visualizations

- [ ] a. are regularly used for visualizing Anscombe's quartet
- [ ] b. represent every data attribute/dimension as a single particle
- [ ] c. are based on sedimentation effects that can be observed in nature
- [x] d. represent every data item as a single particle

Frage **20**

Vollständig

Erreichbare Punkte: 5,00

The following are purposes of explainable AI

- [x] a. teaching & understanding concepts
- [x] b. training models
- [x] c. interpretability & explainability
- [ ] d. applying AutoML approaches

◄ 2022 XAI Lab 05 Assignment 2 Introduction - Recording

Direkt zu: